MULTI-POINT QUASI-FINITE SPLITTING VARIANT: FINITE PIECES WITH PURELY INSEPARABLE RESIDUE TOWERS AT CHOSEN ISOLATED FIBRE POINTS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12135-12159 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 12161-12166, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. BORDERLINE for Chief: proof is an identical-proof pointer between two same-chapter lemmas both outside this lane's collected set (published tags); accepted under the written-reduction rule because the substituted input pair is named exactly, but flagged for possible reclassification/rejection — the substance lives in the referenced base lemma.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The base technical splitting lemma and its single-point etale-makes gadget, plus the variant multi-point gadget (lemma etale-makes-quasi-finite-finite-multiple-points-var, this chapter's preceding card).

Source reference. Original lemma 02LL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11974–11981. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02LM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12019–12026. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02LN: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12094–12101. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $s \in S$. Let $x_1, \ldots, x_n \in X_s$. Assume that*

(1) *f is locally of finite type,*
(2) *f is separated, and*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(3) $x_1, \ldots, x_n$ *are pairwise distinct isolated points of* X_s.

Then there exists an étale neighbourhood $(U, u) \to (S, s)$ *and a decomposition*

$$U \times_S X = W \amalg \coprod\nolimits_{i=1,\ldots,n} \coprod\nolimits_{j=1,\ldots,m_i} V_{i,j}$$

into open and closed subschemes such that the morphisms $V_{i,j} \to U$ *are finite, the fibres of* $V_{i,j} \to U$ *over* u *are singletons* $\{v_{i,j}\}$*, each* $v_{i,j}$ *maps to* x_i*,* $\kappa(u) \subset \kappa(v_{i,j})$ *is purely inseparable, and the fibre of* $W \to U$ *over* u *contains no points mapping to any of the* x_i.

Proof. This is proved in exactly the same way as the proof of Lemma 02LN except that it uses Lemma 02LM instead of Lemma 02LL. □